Do you have sleep problems？

A Machine Learning Study Predicting Sleep Disorders (Insomnia or sleep apnea)

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Data Understanding and EDA

Dataset introduction

- **14 Columns & 374 Rows**
- **Target: Sleep Disorder (None, Insomnia, Sleep Apnea)**
- **Numerical Features:**
  - **'Person ID', 'Age', 'Sleep Duration', 'Quality of Sleep', 'Physical Activity Level', 'Stress Level', 'Heart Rate', 'Daily Steps'**
- **Categorical Features:**
  - **'Gender', 'Occupation', 'BMI Category', 'Blood Pressure', 'Sleep Disorder'**

↓

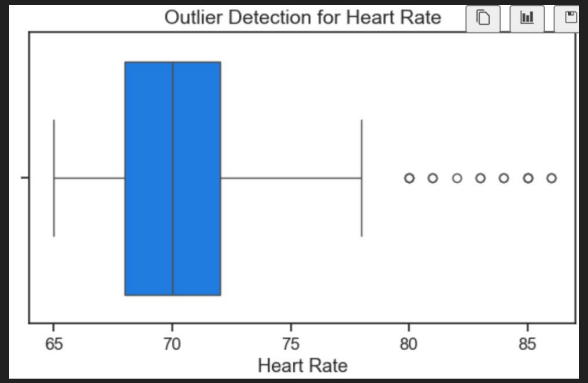
This is a classification problem based on a small dataset

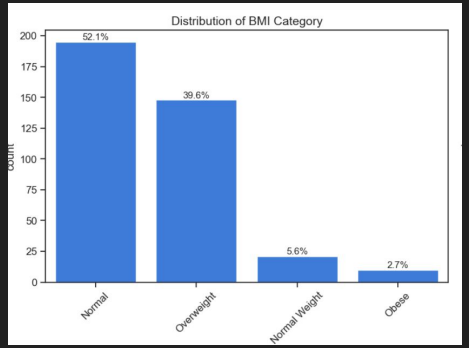
Data Quality Assessment

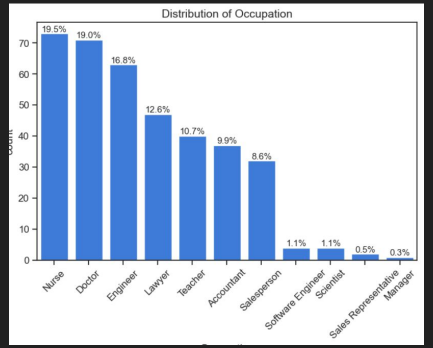
1. **Missing/Duplication:** No missing values or duplicate records were detected.
    a. The 'None' label in 'Sleep Disorder' indicates a healthy status
2. **Data Integrity:** All feature entries are sound and consistent.
3. **Key Transformations Required:**
    a. **Blood Pressure:** can’t be used directly(“123/64”), convert the "Systolic/Diastolic" format into separated numerical or categorical (Hypertension) features.
    b. **BMI Category:** Merge semantically equivalent labels (e.g., "Normal" and "Normal Weight") to ensure consistency.

Univariate Analysis

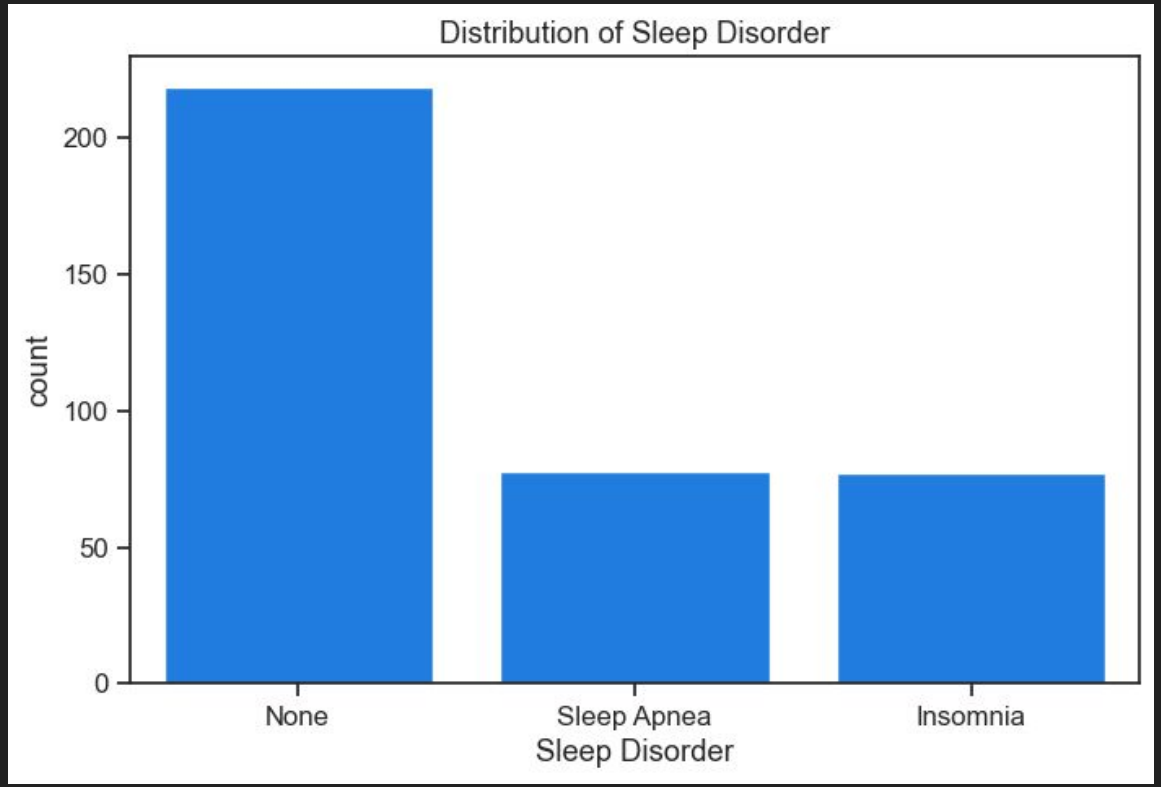
- **Scaling Need Confirmed:** Significant variance in numerical feature scales (e.g., Daily Steps) requires standardization.
- **Outlier Retention:** Clinical outliers (e.g., in Heart Rate) were validated as legitimate observations and retained.
- **Categorical Consolidation:** Semantically similar BMI labels must be merged for data consistency.
- **Sparsity Mitigation:** Low-frequency occupation classes will be grouped to ensure sufficient model pattern learning.







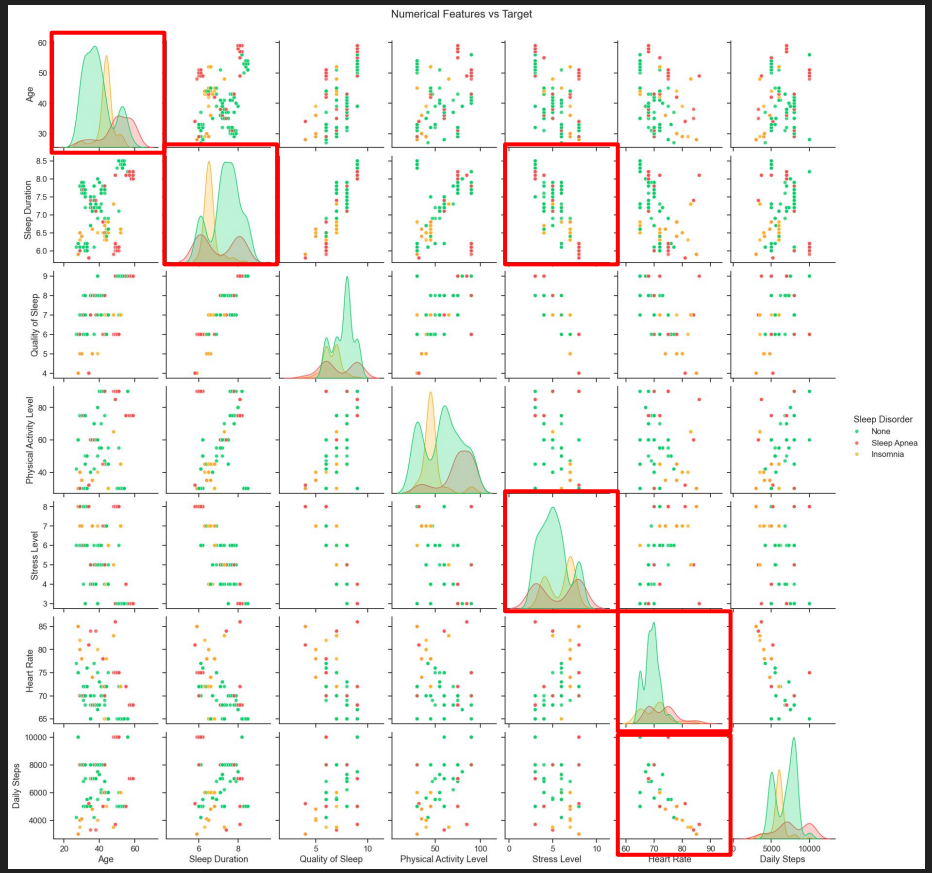
Target Analysis



1. The target variable exhibits class imbalance; therefore, **stratified splitting was applied** to maintain proportional representation in both training and test sets.

2. Sleep Apnea (77 samples) and Insomnia (78 samples) still have sufficient representation for model training. Therefore, oversampling or other resampling techniques are not deemed necessary at this stage.

Numerical Feature & Target

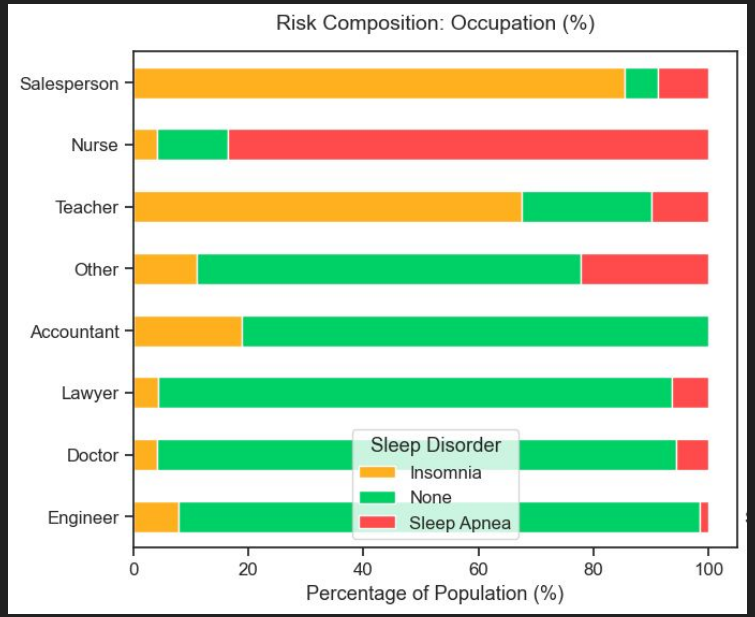


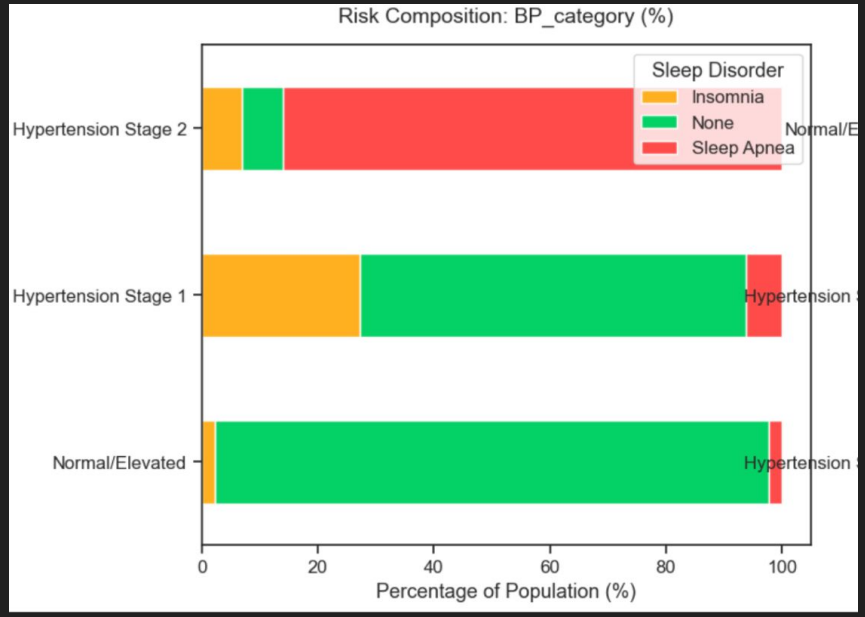
- **Heart Rate Threshold:** A Heart Rate exceeding 80 bpm is a near-certain indicator of a sleep disorder.
- **Age as a Discriminator:** Age effectively differentiates between Healthy, Insomnia, and Sleep Apnea cases, with the prevalence peaks for the three groups concentrating in distinct age brackets.
- **Sleep Duration Insight:** Individuals reporting a sleep duration of approximately 6.5 hours show a higher probability of having Insomnia.
- **Clues for feature engineering**

Categorical Features & Target

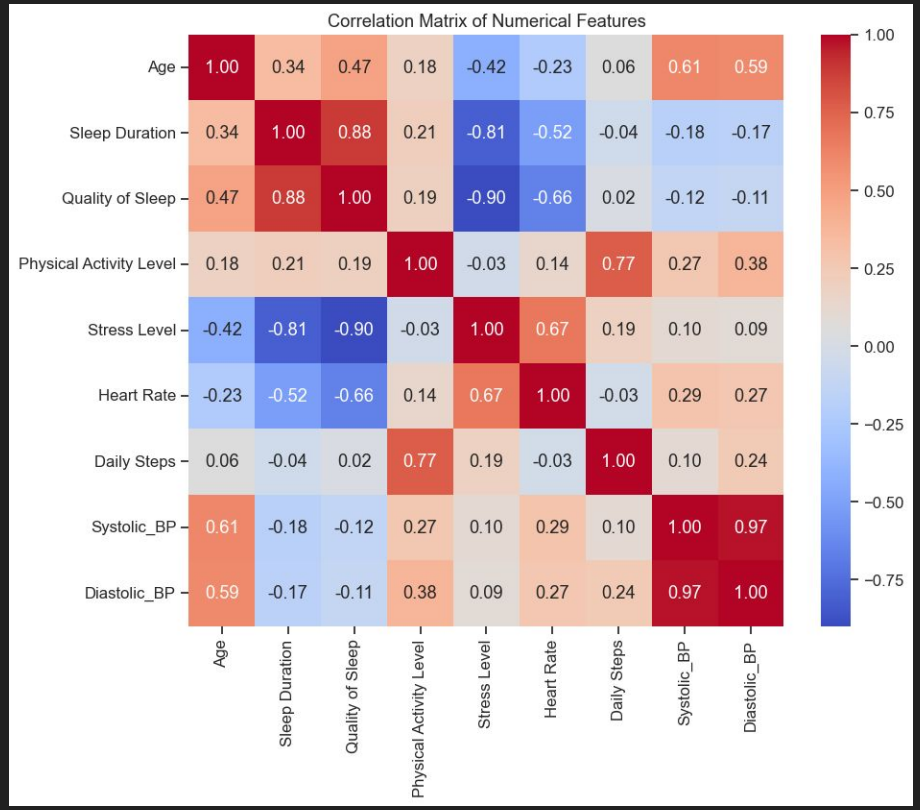
Diagnostic Value: Both **Occupation** and **Blood Pressure metrics** serve as essential and significant indicators for predicting sleep disorders.

- **Occupational Insights:** Significant differences observed. Nurses are predominantly affected by Sleep Apnea, while Salespersons and Teachers show a higher propensity for Insomnia. Other occupations largely belong to the 'Healthy' group.
- **Blood Pressure Correlation:** Stage 2 patients are more susceptible to Sleep Apnea. Stage 1 patients show an elevated probability of Insomnia.

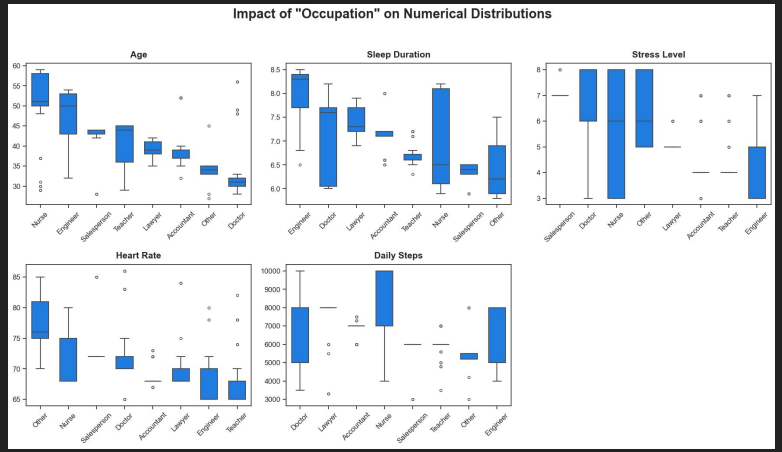




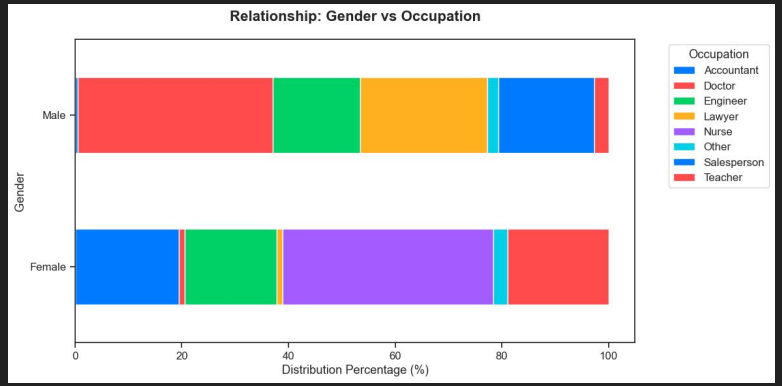
Bivariate Analysis



Numerical Feature Inter-correlation Heatmap



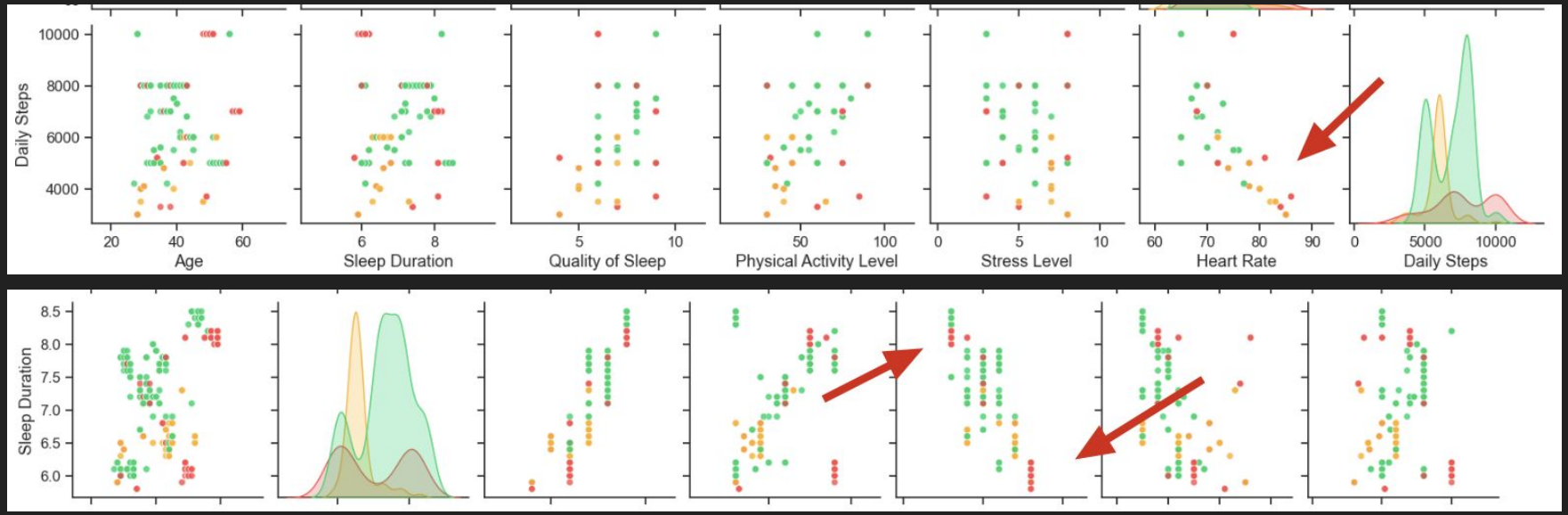
Boxplot（Categorical features vs Numerical features）



Stacked Bar Chart（Categorical vs Categorical features）

Feature Engineering and Data Preprocessing

Feature Engineering

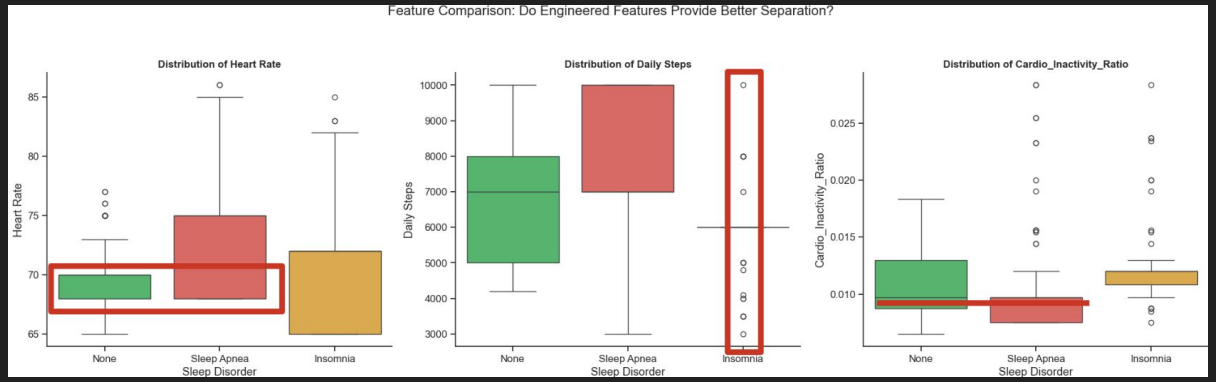


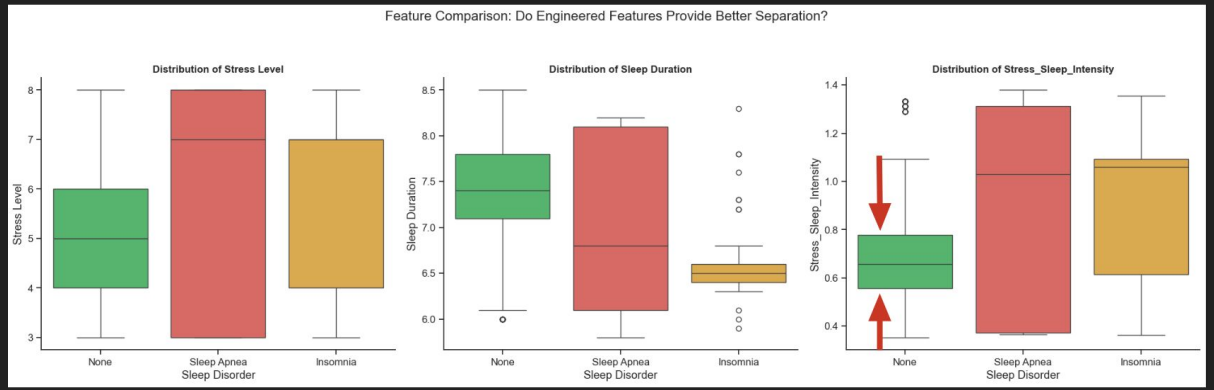
Create two new features

- **Cardio_Inactivity_Ratio : Heart Rate / Daily Steps**, A high heart rate combined with low daily steps results in a large ratio, which allows us to identify periods of cardiovascular inactivity.

- **Stress_Sleep_Intensity : Stress Level / Sleep Duration**, High stress with short sleep, or unusual stress-sleep combinations, generate extreme values that may indicate Sleep Apnea.

Feature Comparison

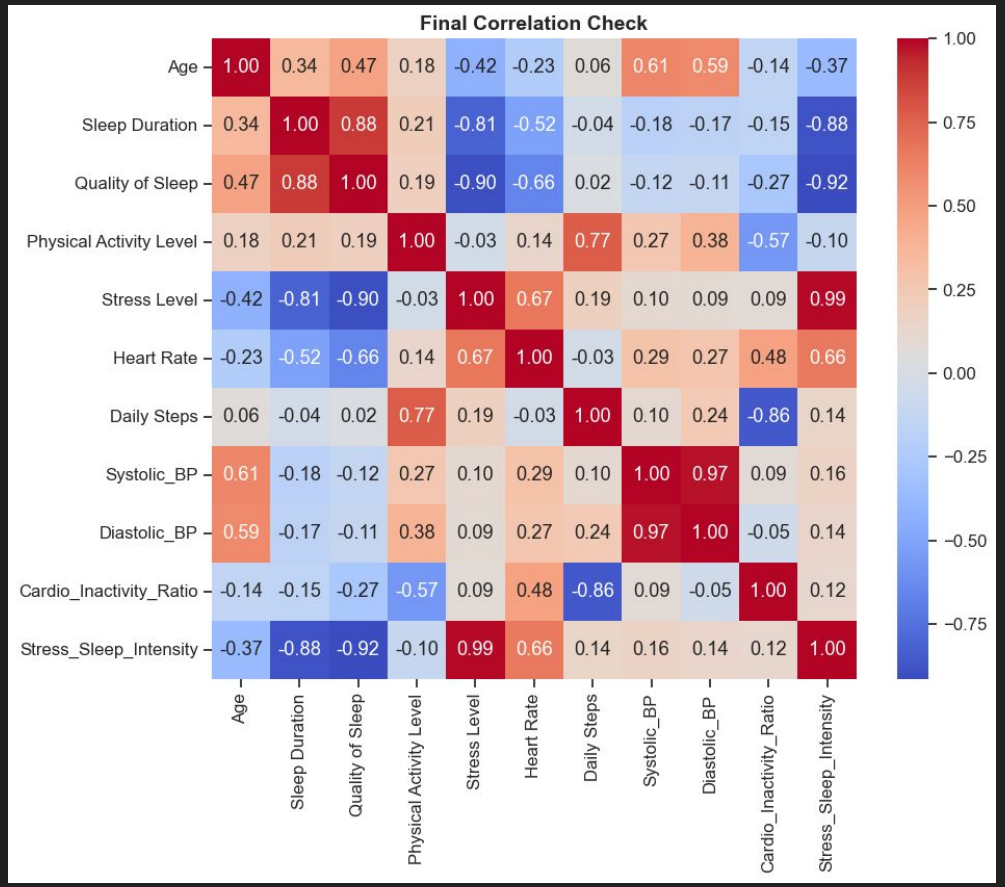




New features outperform the original, with smaller class overlap and better discrimination.

Therefore, it is worth for next step exploring.

Feature Selection



we plan to remove:

- Stress Level(0.99 correlation with Stress_Sleep_Intensity),
- Systolic_BP, Diastolic_BP(BP_category is enough),
- Quality of Sleep(high correlation with Sleep Duration and indication for sleep problems),
- Physical Activity Level(lack of discriminative power)
- Blood Pressure(converted to BP_category)
- Person Id(noise)

Data Preprocessing Workflow

```python
# == 2. TARGET ENCODING ==
# LabelEncoder transforms text labels (None, Insomnia, etc.) into integers (0, 1, 2)
le = LabelEncoder()
y_encoded = le.fit_transform(y)

# == 3. TRAIN-TEST SPLIT ==
# Using an 80/20 split. Stratification ensures the ratio of sleep disorders
# remains the same in both sets to prevent sampling bias.
X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded, test_size=0.20, stratify=y_encoded, random_state=42
)

# NOTE ON DATA SPLITTING STRATEGY:
# 1. No separate manual validation set is created here because we implement
#    5-Fold Cross-Validation during the hyperparameter tuning phase (GridSearchCV).
# 2. This approach maximizes the use of training data, which is crucial given
#    our relatively small dataset (n=374).
# 3. The Test Set (20%) remains completely unseen ('locked') until the final
#    evaluation to provide an unbiased estimate of the model's performance.

# == 4. PREPROCESSING DEFINITION (ColumnTransformer) ==
# Define clinical orders for Ordinal Encoding
bmi_order = ['Normal', 'Overweight', 'Obese']
bp_order = ['Low BP', 'Normal/Elevated', 'Hypertension Stage 1', 'Hypertension Stage 2']

# Group features by their required transformation type
num_features = X.select_dtypes(include=['int64', 'float64']).columns.tolist()
ord_features = ['BMI Category','BP_category']
nom_features = ['Gender', 'Occupation']

# Create the transformer object to handle all feature types in one go
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), num_features),                    # Scale numbers to mean 0, std 1
        ('ord', OrdinalEncoder(categories=[bmi_order, bp_order]), ord_features), # Map order (0<1<2)
        ('nom', OneHotEncoder(handle_unknown='ignore'), nom_features)  # Dummy variables for categories
    ])
```

Dataset Split

- Stratified Train–Test Split
- No validation dataset here because we do cross validation later

Preprocessor

- Encode
  - One Hot Encoder
  - Ordinal Encoder
- Scale
  - StandardScaler

Model Selection and Fine Tuning

Model Selection — Initial Benchmarking

- **Five representative models** from diverse domains were selected for initial benchmarking to assess their performance on this specific dataset.
- **Initial Finding:** Based on the observed Accuracy and F1-Score, the performance of the top three models was found to be **comparable**

```
--- Stage 1 Model Ranking (Screening Results) ---
               Model        Accuracy  Precision (Macro)  Recall (Macro)  F1-Score (Macro)
Logistic Regression 0.8862 ± 0.0290             0.8608          0.8639            0.8606
            XGBoost 0.8830 ± 0.0296             0.8665          0.8560            0.8574
                SVM 0.8763 ± 0.0308             0.8488          0.8554            0.8494
      Random Forest 0.8562 ± 0.0373             0.8370          0.8371            0.8309
                KNN 0.8595 ± 0.0270             0.8352          0.8321            0.8299
```

Hyperparameter fine tuning

```
✅ Logistic Regression Tuning Complete!
Best CV Accuracy: 89.97%
Best Parameters: {'classifier__C': 1.0, 'classifier__penalty': 'l2', 'classifier__solver': 'lbfgs'}
```

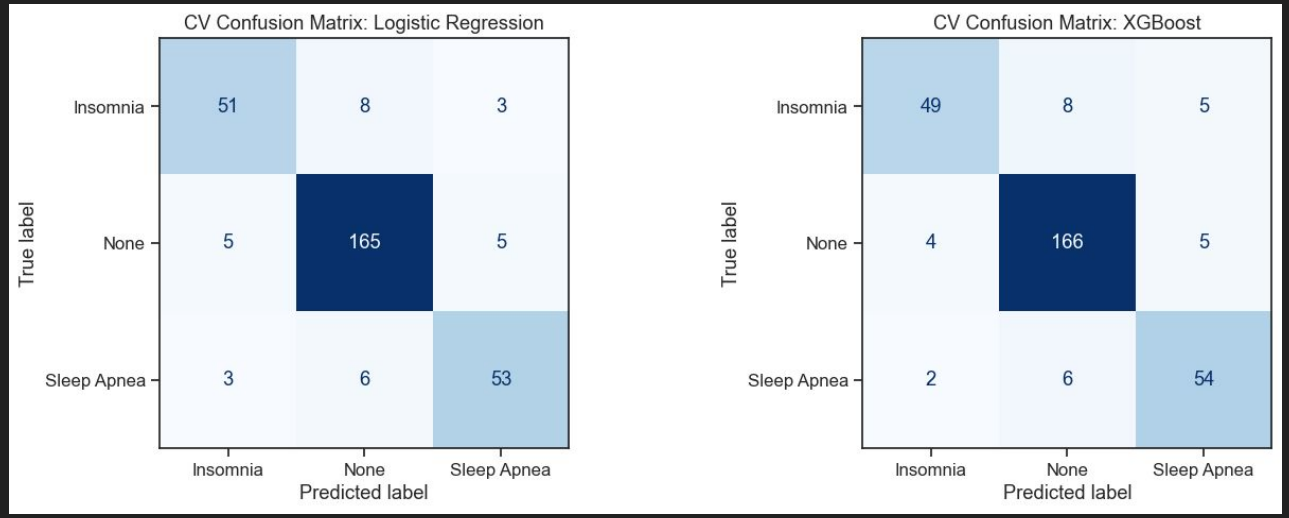
```
⏳ Tuning XGBoost to find the best tree structure...
Fitting 5 folds for each of 54 candidates, totalling 270 fits

✅ XGBoost Tuning Complete!
Best CV Accuracy: 89.97%
Best Parameters: {'classifier__learning_rate': 0.2, 'classifier__max_depth': 3, 'classifier__n_estimators': 50, 'classifier__subsample': 0.8}
```

```
🚀 Tuning SVM for Sleep Disorder Diagnosis...
Fitting 5 folds for each of 48 candidates, totalling 240 fits

✅ SVM Tuning Complete!
Best CV Accuracy: 88.43%
Best Parameters: {'classifier__C': 1, 'classifier__gamma': 'scale', 'classifier__kernel': 'poly'}
```

After hyperparameter tuning, XGBoost **improved from 88.30% to 89.97%**, matching Logistic Regression. We can see that the hyperparameters for LR did not change. Compared to default parameters, **the number of trees, tree depth, and learning rate for XGBoost were all reduced, which helps prevent overfitting on a small dataset**.

Report and Confusion Comparison

```
======================== Logistic Regression ============ ======================== XGBoost ========================
CV-based Classification Report (Internal Validation):      CV-based Classification Report (Internal Validation):
              precision    recall  f1-score   support                  precision    recall  f1-score   support

    Insomnia       0.86      0.82      0.84        62        Insomnia       0.89      0.79      0.84        62
        None       0.92      0.94      0.93       175            None       0.92      0.95      0.94       175
 Sleep Apnea       0.87      0.85      0.86        62     Sleep Apnea       0.84      0.87      0.86        62

    accuracy                           0.90       299        accuracy                           0.90       299
   macro avg       0.89      0.87      0.88       299       macro avg       0.89      0.87      0.88       299
weighted avg       0.90      0.90      0.90       299    weighted avg       0.90      0.90      0.90       299
```



From the training set, both models achieved **90%** accuracy after tuning.

Through the CV Confusion Matrix, XGBoost demonstrates superior performance to Logistic Regression in minimizing missed diagnoses of Sleep Apnea, which is considered a more critical error than missing Insomnia cases.

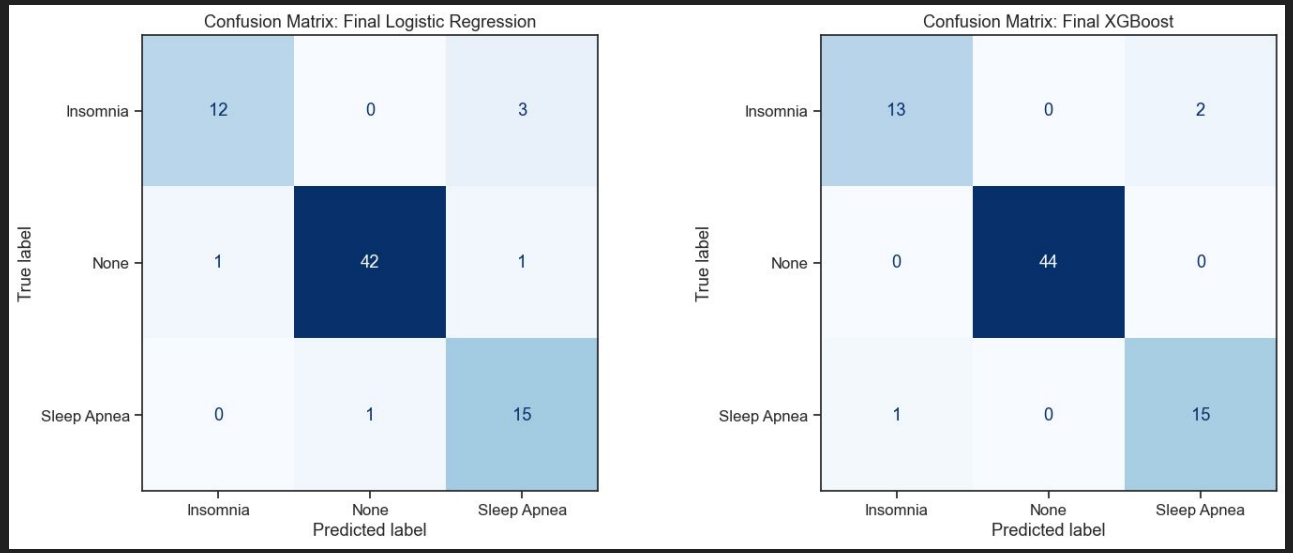
Anyway, let us run on the test validation and take a see first

Final Evaluation and Interpretation

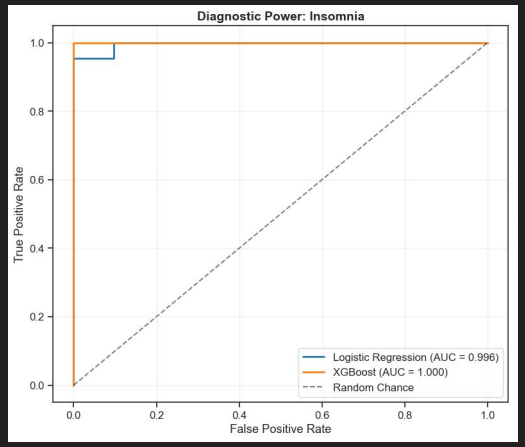
Reports and Confusion Matrix

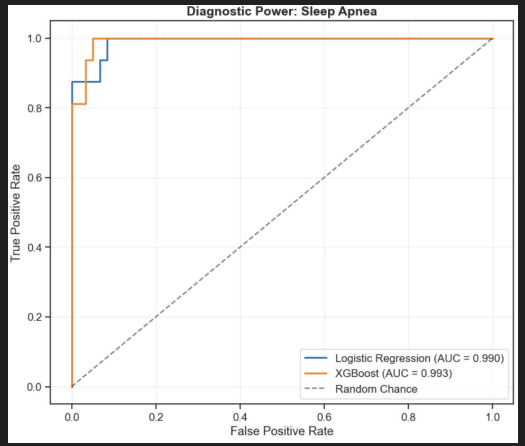
1. XGBoost with **96%** accuracy vs Logistic Regression with **92%** accuracy.
2. XGBoost predicts **all the healthy cases!**
3. XGBoost labels 2 Insomnia wrongly to Sleep Apnea, 1 Sleep Apnea wrongly to Insomnia.

```
======================================
📄 REPORT A: Final Logistic Regression (Test Set)
======================================
              precision    recall  f1-score   support

    Insomnia       0.92      0.80      0.86        15
        None       0.98      0.95      0.97        44
 Sleep Apnea       0.79      0.94      0.86        16

    accuracy                           0.92        75
   macro avg       0.90      0.90      0.89        75
weighted avg       0.93      0.92      0.92        75


======================================
📄 REPORT A: Final XGBoost (Test Set)
======================================
              precision    recall  f1-score   support

    Insomnia       0.93      0.87      0.90        15
        None       1.00      1.00      1.00        44
 Sleep Apnea       0.88      0.94      0.91        16

    accuracy                           0.96        75
   macro avg       0.94      0.93      0.94        75
weighted avg       0.96      0.96      0.96        75
```



ROC and AUC





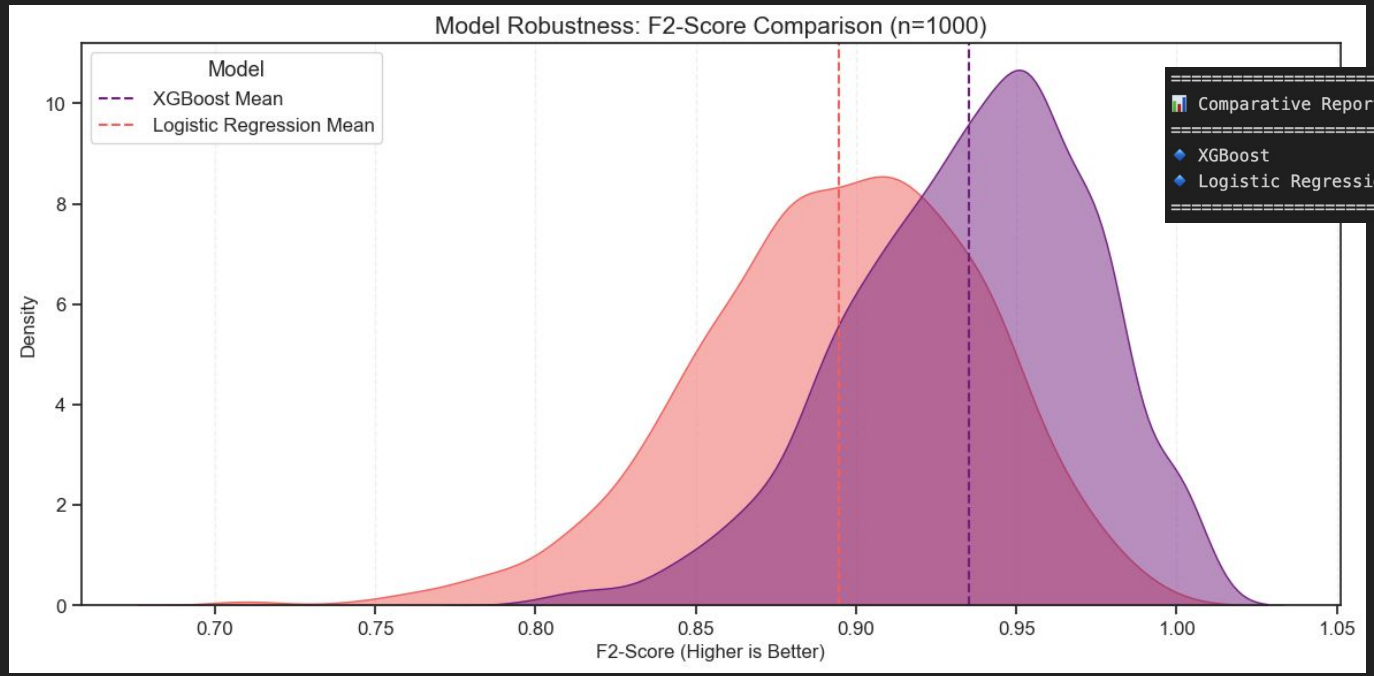
In the medical field, accuracy is not the most crucial metric.**Misdiagnosis and missed diagnosis** are very important observation metrics.

The ROC and AUC curves indicate that both models have **very strong predictive power.**

Especially XGBoost, with an AUC of **100%** for predicting Insomnia and **99.3%** for predicting Sleep Apnea. This shows model could distinguish one disease from the other two.

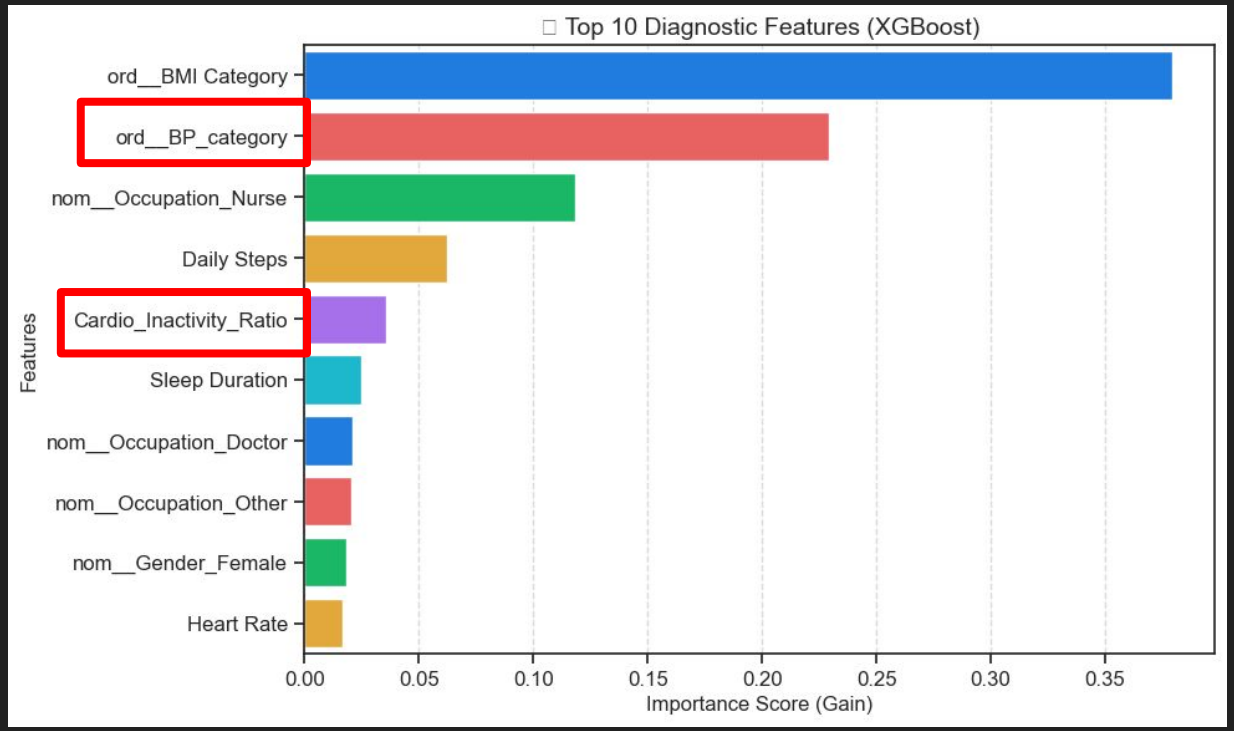
Bootstrapping：Luck or Strength？

1. **XGBoost > Logistic Regression** (Mean F2: 0.935 vs 0.895)
2. **XGBoost robust:** F2 ≥ 0.853 in 95% of resamples




```
================================================
📊 Comparative Report: F2-Score (95% CI)
================================================
🔹 XGBoost              | Mean: 0.9352 | 95% CI: [0.8528, 1.0000]
🔹 Logistic Regression  | Mean: 0.8945 | 95% CI: [0.8008, 0.9704]
================================================
```

Which features contribute



The analysis confirms that BP_category and Cardio_Inacticity_Ratio are among the top 5 most contributing features.

Thank You